

Proof ledger

Repository source: proof/PROOF_LEDGER.md

Status labels:

- **Sourced:** stated and proved in an identified reference.
- **Hand-checked:** algebra reviewed in the project notes.
- **Exact-code checked:** a symbolic identity was verified over general symbolic parameters.
- **Numerically cross-checked:** deterministic quadrature/random tests agree.
- **Formal write-up drafted:** a line-by-line proof is in the repository, but has not received outside review.
- **Integrated manuscript drafted:** the claim has been merged into the LaTeX research note and compiled.
- **CI configured:** a hosted workflow has been added but its first repository run still needs confirmation.
- **Needs local rerun:** a matching check exists but should be rerun and preserved in the current repository state.
- **Needs external review:** no independent subject-matter expert has checked it.

ID	Claim	Current evidence	Status
N1	Support-function constraints and volume formula	Nishioka, Sections 2–4	Sourced
N2	Smooth approximation and volume continuity	Nishioka, Lemma 2; Martini–Montejano–Oliveros, Theorem 8.5.1; limiting argument in BRIDGE_LEMMAS.md	Sourced; formal write-up drafted
S1	Degree-3 coefficient is $1/22$; tail coefficient is at most $1/58$	Nishioka, Lemma 4 and Proposition 1; explicit derivation in BRIDGE_LEMMAS.md	Sourced; formal write-up drafted
S2	Tensor quadratic form is orthogonal across harmonic degrees	Polarized Bochner calculation in BRIDGE_LEMMAS.md, Lemmas 2.1–2.2	Hand-checked; formal write-up drafted; needs external review
T1	General cubic tensor is harmonic iff symmetric trace-free	DEGREE3_TENSOR_IDENTITY.md, Section 2; symbolic Laplacian check	Exact-code checked; formal write-up drafted; needs external review
T2	$ A_Y _2^2 = (176\pi/7)\tau$	Moment derivation in DEGREE3_TENSOR_IDENTITY.md, Section 5; general symbolic integration	Exact-code checked; formal write-up drafted; needs external review
T3	Exact formula for $ 4 \det A_Y _2^2$	Quartic contraction derivation in DEGREE3_TENSOR_IDENTITY.md, Sections 6–8; general symbolic integration	Exact-code checked by two formulations; formal write-up drafted; needs external review
T4	Compact pointwise determinant formula	DEGREE3_TENSOR_IDENTITY.md, Sections 3–4; matrix formula compared modulo $x^2+y^2+z^2-1$	Exact-code checked; hand-checked; needs external review

ID	Claim	Current evidence	Status
T5	Rotational invariance and random convention checks	Deterministic sphere quadrature and random rotations	Numerically cross-checked independently in Python and R; both earlier runs pass
T6	Pairing counts for the six quartic moments	Hand count in <code>DEGREE3_TENSOR_IDENTITY.md</code> , Section 7; exhaustive exact pairing enumeration	Python exact-count check passed; matching R check included; needs local rerun/preserved output in current state
F1	Nuclear-norm identity and Cauchy–Schwarz step	<code>BRIDGE_LEMMAS.md</code> , Lemmas 4.1–4.2	Hand-checked; formal write-up drafted; needs external review
F2	Projection/duality step $\ C\ _2^2 = \langle A_h, C \rangle \leq (d/2) \int \ C\ _*$	<code>BRIDGE_LEMMAS.md</code> , Corollary 2.3 and Lemmas 5.1–5.2	Hand-checked; formal write-up drafted; needs external review
A1	Final exact coefficient, decimal, and strict improvement	SymPy exact arithmetic; <code>BRIDGE_LEMMAS.md</code> , Theorem 6.2 and Corollary 6.3	Exact-code checked; hand-checked; integrated manuscript drafted
R1	Passage to nonsmooth bodies	Nishioka’s approximation lemma; <code>BRIDGE_LEMMAS.md</code> , Theorem 7.1	Sourced; formal write-up drafted; needs external review
P1	Complete paper source compiles without unresolved references	<code>paper/degree3_spectral_refinement.tex</code> ; <code>build_paper.sh</code>	Integrated manuscript drafted; runtime-tested in project environment
P2	PDF has no visible clipping, overlap, or broken glyphs	Eight rendered pages inspected at 180 dpi	Render-checked
C1	Hosted clean-environment checks for Python, R, the paper, and Markdown PDFs	<code>.github/workflows/verification.yml</code>	CI configured; first GitHub-hosted run must still be confirmed
D1	Public-facing verification and reproduction instructions	<code>VERIFICATION_STATUS.md</code> ; <code>REPRODUCIBILITY.md</code>	Drafted and integrated
L1	Exact constant/decimal not found in a targeted web search	Search terms and limitations recorded in <code>PROOF_AUDIT.md</code>	Preliminary only; not a systematic novelty review
E1	Full theorem has been independently reviewed	None yet	Needs external review

Current bottleneck

The internal proof chain and first integrated manuscript are complete. Continuous integration and verification documentation are now configured. The next internal gates are:

1. confirm that the first remote GitHub Actions run passes all three jobs;
2. have the named author finish reading and approve the manuscript and disclosure language;
3. choose the public license and finalize `CITATION.cff`;
4. perform and record a fresh-clone test before public release.

Release rule

Do not describe the candidate bound as peer reviewed, certified, or independently verified while **E1** remains open. The first public GitHub release may be labeled **public research draft seeking verification** after all internal gates above pass.